PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

Construção de mapas de Hi-C

Amanda Souza Câmara

Leibniz Institute of Plant Genetics and Crop Plant Research

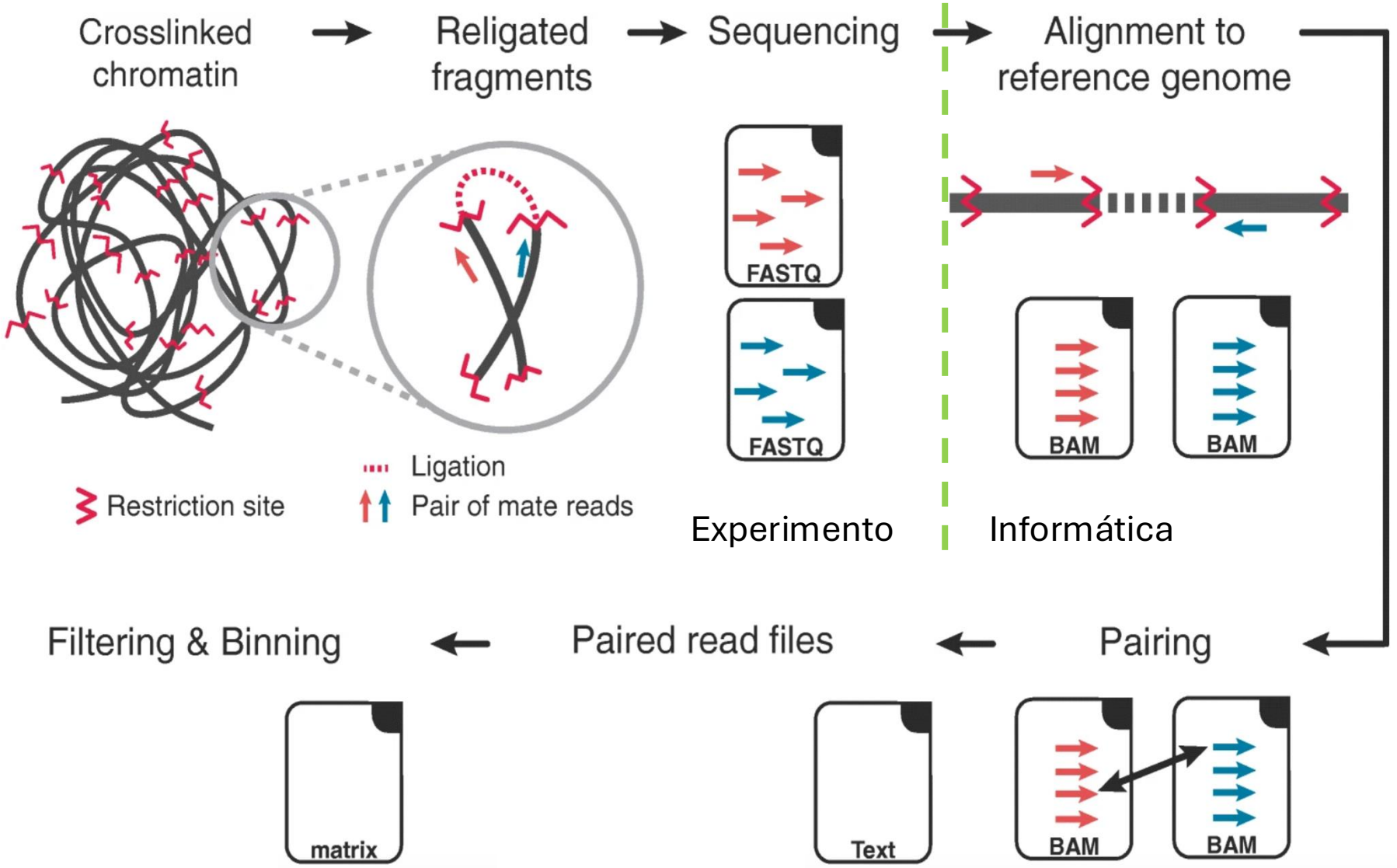


26 de outubro de 2026

In Silico Genebank Proteomics

IPK LEIBNIZ-INSTITUT

Como construir o mapa de contatos



Pal et al, 2018

Como construir o mapa de contatos

- Ler os pares de sequências de Hi-C
- Alinhar no genoma de referênca
- Identificar sítios pares
- Popular a matriz de pares sítios com ocorrências de pares observados

Muitos pipelines diferentes

- ICE (Imakaev et al. 2012),
- TADbit (Serra et al. 2017)
- HiCUP (Wingett et al. 2015)
- HIPPIE (Hwang et al. 2015)
- Juicer (Durand et al. 2016b)
- HiC-Pro (Servant et al. 2015)
- HiCExplorer (Ramirez et al. 2018)
- Ferramentas independentes combinadas (samtools, cutadapt,bwa, minimap,...)

Juicer

Author Manuscript
Peer reviewed and accepted for publication by a journal



Author Manuscript

Cell Syst. Author manuscript; available in PMC: 2018 Mar 12.

Published in final edited form as: Cell Syst. 2016 Jul;3(1):95–98. doi: 10.1016/j.cels.2016.07.002

Juicer provides a one-click system for analyzing loop-resolution Hi-C experiments

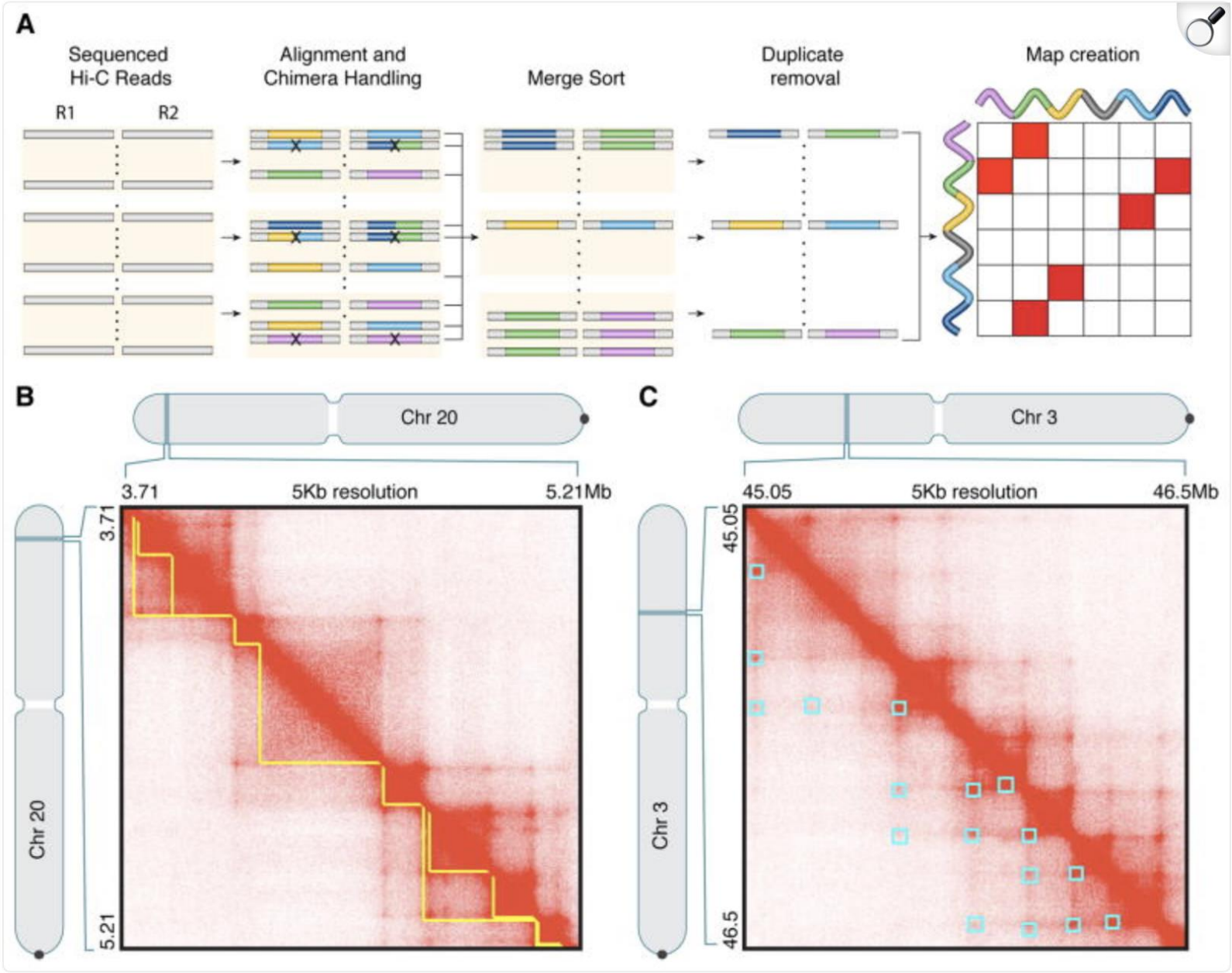
Neva C Durand [1,2,3,4,*], Muhammad S Shamim [1,2,3,*], Ido Machol [1,2,3], Suhas S P Rao [1,2,3,5], Miriam H Huntley [1,2,3,6], Eric S Lander [4,7,8], Erez Lieberman Aiden [1,2,3,4,9]

Author information Copyright and License information

PMCID: PMC5846465 NIHMSID: NIHMS804301 PMID: 27467249

Juicer

Figure 1. Juicer analyzes terabases of Hi-C data with one click.



Tudo com juicer

- Ler os pares de sequências de Hi-C
- Alinhar no genoma de referênca
- Identificar sítios pares
- Popular a matriz de pares sítios com ocorrências de pares observados

Mas é preciso preparar a bancada

Instalando dependências

- Python normalmente já está instalado

- Bwa, samtools e java no mac:
- brew install samtools
- brew install bwa
- brew install openjdk

- No linux:
- sudo apt update
- sudo apt-get install samtools
- sudo apt install bwa
- sudo apt install default-jdk

Organizar pastas (diretórios) e arquivos

- Preparar o juicer para encontrar os arquivos nos locais certos
- Ler os pares de sequências de Hi-C
  - Preparar arquivos fastq R1 e R2
- Alinhar no genoma de referênca
  - Preparar o genoma para ser lido
  - Encontrar os possíveis sítios de digestão
- Identificar sítios pareados
- Popular a matriz de pares de sítios com ocorrências de pares observados

Preparando o juicer

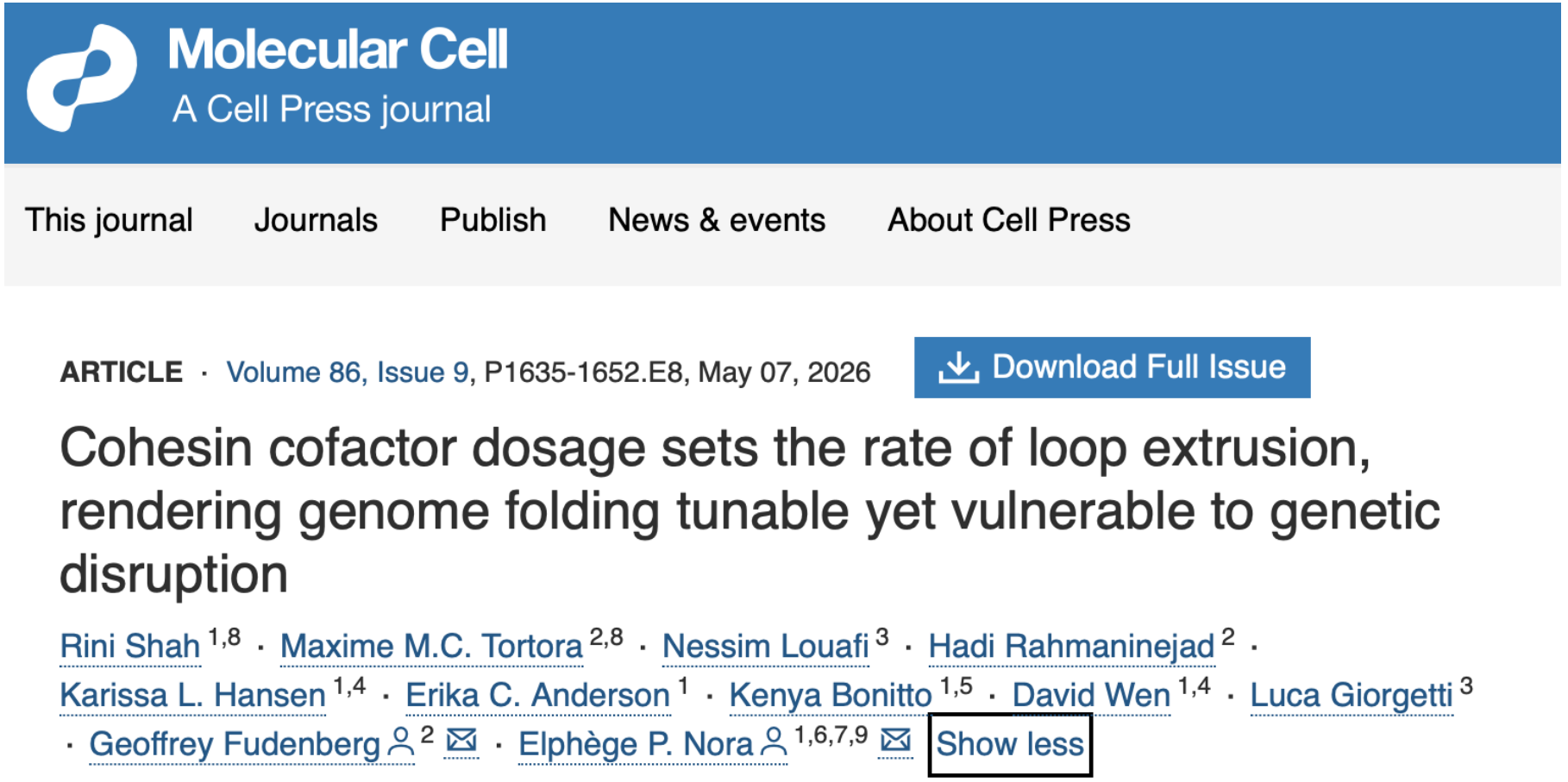
- Crie uma pasta para as análises do curso, dentro dela:
- git clone https://github.com/theaidenlab/juicer.git # baixar o juicer do github
- mkdir myjuicer
- cd myjuicer
- ln -s ../juicer/CPU scripts # copia scripts do juicer para sua propria pasta
- cd scripts/common
- wget https://hicfiles.tc4ga.com/public/juicer/juicer_tools.1.9.9_jcuda.0.8.jar # baixar o pacote de ferramentas do juicer
- ln -s juicer_tools.1.9.9_jcuda.0.8.jar juicer_tools.jar
- cd ../.. # volta para myjuicer
- mkdir references
- mkdir restrictions_sites

- Só para mac, substituir comando awk por gawk no script principal do juicer
- brew install gawk
- **alias** awk=/opt/homebrew/bin/gawk
- sed -i '' '545s/awk /gawk /;547s/awk /gawk /;552s/awk /gawk /;554s/awk /gawk /' scripts/juicer.sh
- sed -i '' 's/| awk -f \$juiceDir\/scripts\/common\/dups_sam.awk/| gawk -f $juiceDir\/scripts\/common\/dups_sam.awk/g' scripts/juicer.sh

Input

- Arquivos de pares de sequências de Hi-C, fastq (x2)
- Sequência do genoma de referência, fasta
- Informação sobre enzimas de digestão usadas

Nosso exemplo



https://www.cell.com/molecular-cell/abstract/S1097-2765(26)00233-9

Encontrar informações de entrada

- Onde estão guardados os arquivos de pares de sequências de Hi-C (fastq)
  - Normalmente no Sequence Read Archive (SRA da NCBI, criado em 2007, https://en.wikipedia.org/wiki/Sequence_Read_Archive)

- Qual foi o genoma de referência e onde encontrá-lo

- Qual foi a enzima de digestão usada?

Organizando dados de entrada

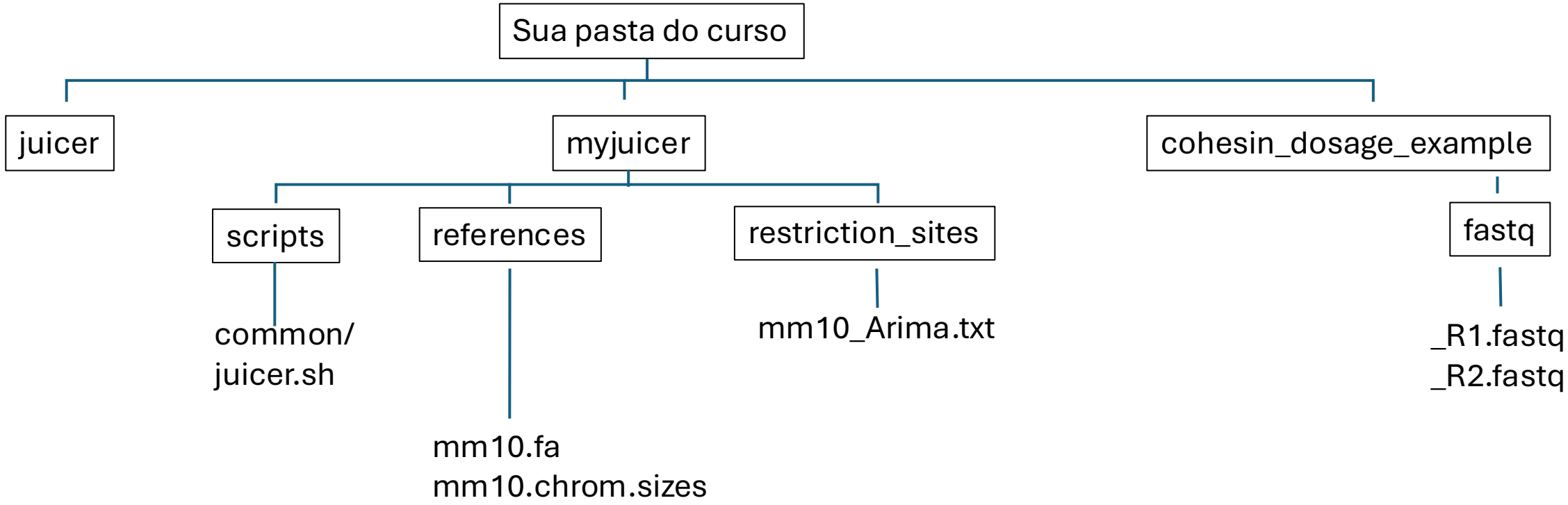
- Na pasta myjuicer
- Baixar o toolkit do sra para baixar arquivos fastq com uma das duas opções
  - curl -L -o sratoolkit.tar.gz https://ftp-trace.ncbi.nlm.nih.gov/sra/sdk/current/sratoolkit.current-mac-arm64.tar.gz
  - curl -L -o sratoolkit.tar.gz https://ftp-trace.ncbi.nlm.nih.gov/sra/sdk/current/sratoolkit.current-alma_linux64.tar.gztar -xvzf sratoolkit.tar.gz
- tar -xvzf sratoolkit.tar.gz
  - **export** PATH="$PWD/sratoolkit.3.4.1-alma_linux64/bin:$PATH”
  - **export** PATH="$PWD/sratoolkit.3.4.1-mac-arm64/bin:$PATH”

- Na sua pasta do curso
- mkdir cohesin_dosage_example
- cd cohesin_dosage_example
- mkdir fastq
- cd fastq
- prefetch SRR30953179
- fasterq-dump --split-files SRR30953179
- # mudar nome dos arquivos para terminar em _R1.fastq e _R2.fastq

Preparando o genoma

- Na pasta myjuicer
- cd references
- curl -O https://hgdownload.soe.ucsc.edu/goldenPath/mm10/bigZips/mm10.fa.gz (o genoma de mus musculus, mm10)
- gunzip mm10.fa.gz
- samtools faidx mm10.fa #para recuperar o tamanho dos cromossomos
- cut -f1,2 mm10.fa.fai > mm10.chrom.sizes
- bwa index mm10.fa
- cd ../restrictions_sites
- chmod +x ../../juicer/misc/generate_site_positions.py
- ../../juicer/misc/generate_site_positions.py Arima mm10 ../references/mm10.fa

Esquema de pastas e arquivos para rodar o juicer.sh



Rodar o juicer

- Na pasta cohesion_dosage_example
- ../myjuicer/scripts/juicer.sh -t 8 -D ../myjuicer/ -z ../myjuicer/references/mm10.fa -p ../myjuicer/references/mm10.chrom.sizes -y ../myjuicer/restrictions_sites/mm10_Arima.txt -s Arima &

Estatísticas dos reads

- Copiar hic_bam_summary.sh do github do curso para sua pasta do curso
- Na pasta cohesin_dosage_example
- bash ../hic_bam_summary.sh aligned/merged_dedup.bam cohesin_dosage_example

Column	Meaning
sample	Sample name supplied when running the script.
bam_pairs	Number of complete primary read pairs represented in merged_dedup.bam. This is not the original number of raw FASTQ pairs.
duplicate_pairs	Number of read pairs for which at least one mate has the BAM duplicate flag (0x400).
duplicate_fraction	duplicate_pairs / bam_pairs. Fraction of pairs in the BAM marked as duplicates.
both_mapped_pairs	Number of pairs for which both mates mapped to the reference. Includes duplicates and pairs with low MAPQ.
singleton_pairs	Number of pairs for which exactly one mate mapped.
singleton_fraction	singleton_pairs / bam_pairs. Fraction of pairs with only one mapped mate.
non_dup_q10	Number of pairs passing those filters.
p_non_dup_q10	non_dup_q10 / bam_pairs. Fraction of BAM pairs retained after these filters.
pairs_ge_1kb	Number of filtered pairs that are either interchromosomal or map at least 1 kb apart on the same chromosome.
p_ge_1kb	pairs_ge_1kb / non_dup_q10. Fraction of filtered pairs meeting the ≥1-kb definition.
pairs_ge_1mb	Number of filtered pairs mapping to the same chromosome at least 1 Mb apart.
p_ge_1mb_among_ge_1kb	pairs_ge_1mb / pairs_ge_1kb. Fraction of the ≥1-kb group that is intrachromosomal and separated by at least 1 Mb.
interchromosomal	Number of filtered pairs whose mates map to different chromosomes.
p_interchromosomal_among_ge_1kb	interchromosomal / pairs_ge_1kb. Fraction of the ≥1-kb group that is interchromosomal.

*Feito com IA

Para visualizar

- Instalar o Juicebox
- Abrir cohesion_dosage_example/aligned/inter.hic

./juicer.sh -h

1. Reference genome and input parameters

*Feito com IA

Option	Meaning	Example
-g	Genome ID supported by the script	-g hg38
-d	Top-level experiment directory containing fastq/; Juicer creates splits/ and aligned/here	-d /path/to/sample
-z	Reference genome FASTA	-z references/TAIR10.fa
-p	Chromosome sizes file, defining which chromosomes go into the .hic file	-p references/TAIR10.chrom.sizes
-y	Restriction-site positions file for the reference assembly	-y restriction_sites/TAIR10_HindIII.txt
-s	Restriction enzyme/site setting	-s HindIII
-b	Ligation-junction sequence, used when counting ligation junctions	-b GATCGATC
-D	Parent directory of the Juicer installation, containing scripts/, references/ and restriction_sites/	-D /path/to/myjuicer

./juicer.sh -h

2. Threads and metadata

*Feito com IA

Option	Meaning	When to change it
-t	Number of threads for BWA align ment	Increase when CPU resources are available
-T	Number of threads for .hic file cre ation	Increase if the file-creation stage is slow and resour ces allow
-a	Experiment description stored in t he output metadata/statistics	Useful for recording treatment, tis sue or experimental condition
-i	Sample name in the BAM read-group metadata	Set to the biological sample nam e
-k	Library name in the read-group metadata	Set to the library identifier

./juicer.sh -h

3. Duplicate handling and output options

*Feito com IA

Option	Meaning	Practical advice
-w	Wobble distance used in duplicate detection; the CPU script's default is 4	Keep the default unless you have a reason to change it
-j	Use exact duplicate matching instead of wobble-based matching	Changes duplicate-detection behavior; do not change casually
-f	Include fragment-delimited maps in .hic creation	Use if you need restriction-fragment-resolution maps
-u	Treat the data as single-end rather than paired-end	Ordinary paired-end Hi-C generally does not use this
-m	Methylation-library mode	Only for the relevant specialized workflow
-e	Exit early before final .hic creation	Useful in specialized workflows, but not for producing the normal final output

./juicer.sh -h

*Feito com IA

4. Restarting a failed run with -S

-S value	Intended restart point
chimeric	Alignment has completed; resume chimeric-read handling
merge	Chimeric handling has completed; resume merging
dedup	Merged alignments exist; resume duplicate handlin g
final	Deduplication is complete; create final statistics an d .hic files
postproc	.hic files exist; resume feature-annotation post-processing
early	Stop before normal final .hic creation

./juicer.sh -h

5. Other options

*Feito com IA

Option	Purpose
--assembly	Early- exit workflow that creates the older merged_nodup s output for downstream assembly workflows
--cleanup	Automatically clean up intermediate files after succ essful completion
--qc	QC- oriented run mode with a reduced resolution range, according to the script help
--qc_apa	Run QC Aggregate Peak Analysis
--in-situ	Specialized mode for in-situ Hi-C processing
-h or --help	Display help

Nosso exemplo

JOURNAL ARTICLE

Chromatin remodeling complexes regulate genome architecture in Arabidopsis FREE

Tingting Yang, Dingyue Wang, Guangmei Tian, Linhua Sun, Minqi Yang, Xiaochang Yin, Jun Xiao, Yu Sheng, Danmeng Zhu, Hang He ... Show more

Author Notes

The Plant Cell, Volume 34, Issue 7, July 2022, Pages 2638–2651,

https://doi.org/10.1093/plcell/koac117

Published: 21 April 2022 **Article history** ▾

https://academic.oup.com/plcell/article/34/7/2638/6571821?login=false#365062977

Organizando dados de entrada

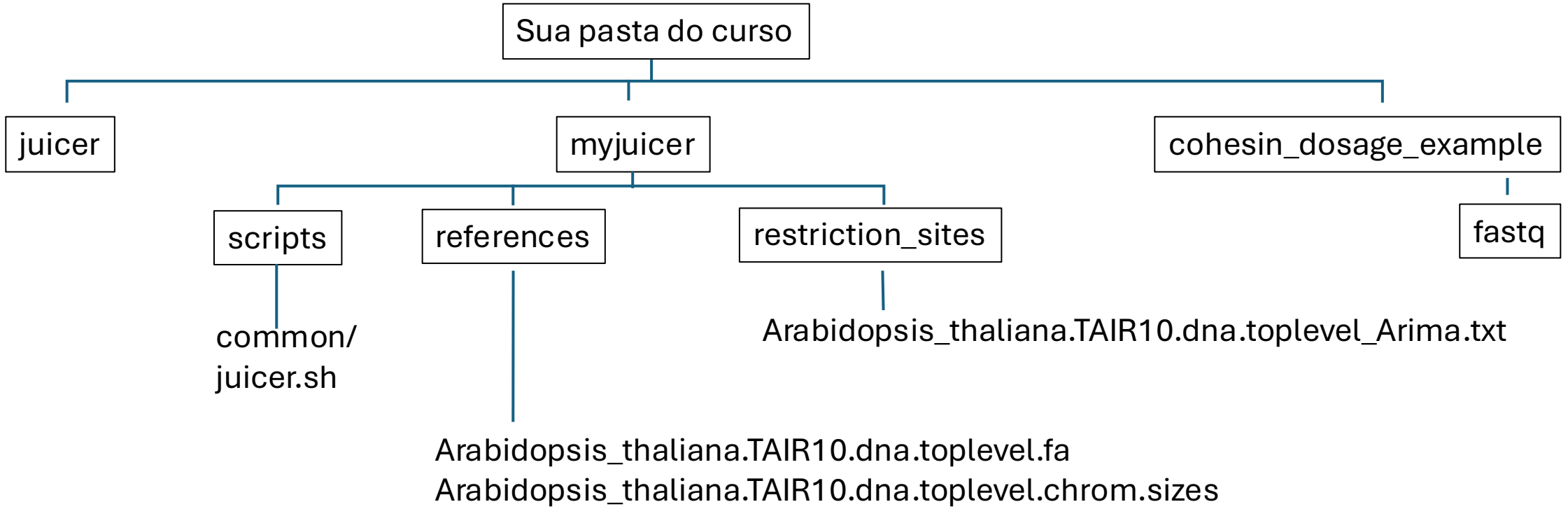
- Na pasta myjuicer
- Baixar o toolkit do sra para baixar arquivos fastq com uma das duas opções
  - curl -L -o sratoolkit.tar.gz https://ftp-trace.ncbi.nlm.nih.gov/sra/sdk/current/sratoolkit.current-mac-arm64.tar.gz
  - curl -L -o sratoolkit.tar.gz https://ftp-trace.ncbi.nlm.nih.gov/sra/sdk/current/sratoolkit.current-alma_linux64.tar.gztar -xvzf sratoolkit.tar.gz
- tar -xvzf sratoolkit.tar.gz
  - **export** PATH="$PWD/sratoolkit.3.4.1-alma_linux64/bin:$PATH”
  - **export** PATH="$PWD/sratoolkit.3.4.1-mac-arm64/bin:$PATH”

- Na sua pasta do curso
- mkdir yang_arabidopsis_example
- cd yang_arabidopsis_example
- mkdir fastq
- cd fastq
- prefetch PRJNA780072
- fasterq-dump --split-files PRJNA780072
- # mudar nome dos arquivos para terminar em _R1.fastq e _R2.fastq

Preparando o genoma

- Na pasta myjuicer
- cd references
- wget https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-63/fasta/arabidopsis_thaliana/dna/Arabidopsis_thaliana.TAIR10.dna.toplevel.fa.gz
- gunzip Arabidopsis_thaliana.TAIR10.dna.toplevel.fa.gz
- samtools faidx Arabidopsis_thaliana.TAIR10.dna.toplevel.fa #para recuperar o tamanho dos cromossomos
- cut -f1,2 Arabidopsis_thaliana.TAIR10.dna.toplevel.fa.fai > Arabidopsis_thaliana.TAIR10.dna.toplevel.chrom.sizes
- bwa index Arabidopsis_thaliana.TAIR10.dna.toplevel.fa
- cd ../restrictions_sites
- chmod +x ../../juicer/misc/generate_site_positions.py
- ../../juicer/misc/generate_site_positions.py Arabidopsis_thaliana.TAIR10.dna.toplevel Arabidopsis_thaliana.TAIR10.dna.toplevel.fa

Esquema de pastas e arquivos para rodar o juicer.sh



SRR16937571_R1.fastq
SRR16937571_R2.fastq